Recap: Fourier series

- If $f(x)$ is piecewise smooth on $(-L, L]$ then its Fourier series is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x), \qquad k_n = \frac{n\pi}{L}$$

  where $f(x)$ is continuous

- The coefficients are given by Euler-Fourier formulae
- We denote the periodic extension of $f(x)$ as $F(x)$
- If $F(x)$ continuous then Fourier series convergence is uniform (e.g., $f = x^2$)
- If $F(x)$ is discontinuous then Gibbs phenomenon occurs (e.g., $f = x$)

In Lecture 18, we will:

1. Define complex Fourier series
2. Define even/odd periodic extensions of functions and calculate their Fourier series

Complex exponentials create a more compact Fourier series

Definition

- The function $f(x)$ on $(-L, L]$ has the *complex Fourier series*

$$\sum_{n=-\infty}^{\infty} c_n \mathrm{e}^{\mathrm{i}k_n x} \quad \text{where} \quad c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\mathrm{e}^{-\mathrm{i}k_n x}\mathrm{d}x$$

- This will be our starting point for Fourier transforms (Topic 6)

Exercise 4.6: Calculate the complex Fourier series of $f(x) = x$ on $(-\pi, \pi]$

Recall that for $f(x) = x$ with $L = \pi$ and $k_n = n$

$$F(x) = -2\sum_{n=1}^{\infty} \frac{(-1)^n}{n}\sin(nx)$$

where it is continuous.

For $n \neq 0$

$$2\pi\, c_n = \int_{-\pi}^{\pi} x\, \mathrm{e}^{-\mathrm{i}nx}\, \mathrm{d}x = \left[\frac{x\, \mathrm{e}^{-\mathrm{i}nx}}{-\mathrm{i}n}\right]_{-\pi}^{\pi} + \frac{1}{\mathrm{i}n}\int_{-\pi}^{\pi} \mathrm{e}^{-\mathrm{i}nx}\, \mathrm{d}x$$

$$= \frac{2\pi(-1)^n}{-\mathrm{i}n} + \left[\frac{\mathrm{e}^{-\mathrm{i}nx}}{n^2}\right]_{-\pi}^{\pi} = \frac{2\pi\mathrm{i}(-1)^n}{n}$$

$$\text{and} \qquad 2\pi\, c_0 = \int_{-\pi}^{\pi} x\, \mathrm{d}x = 0$$

Therefore,

$$F(x) = \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{\mathrm{i}(-1)^n}{n}\mathrm{e}^{\mathrm{i}nx} \qquad (\text{where } F \text{ cts})$$

$$= \sum_{n=1}^{\infty} \frac{\mathrm{i}(-1)^n}{n}\left(\mathrm{e}^{\mathrm{i}nx} - \mathrm{e}^{-\mathrm{i}nx}\right) = -2\sum_{n=1}^{\infty} \frac{(-1)^n}{n}\sin(nx)$$

Fourier series of even and odd functions

Consider the Fourier series of $f(x)$ on $(-L, L]$, i.e.,

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty}\Big[a_n \cos(k_n x) + b_n \sin(k_n x)\Big] \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x \quad \text{and} \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x$$

Case f odd, i.e., $f(-x) = -f(x)$

The Fourier series simplifies to

$$\sum_{n=1}^{\infty} b_n \sin(k_n x) \quad \text{where} \quad b_n = \frac{2}{L}\int_{0}^{L} f(x)\sin(k_n x)\mathrm{d}x$$

Even and odd extensions

- Let $f(x)$ be defined on $x \in [0, L]$
- We can calculate its *even periodic extension* $F(x)$, such that

$$\begin{aligned} F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\ F(x) &= f(-x) \quad \text{for} \quad x \in (-L, 0) \\ F(x + 2L) &= F(x) \end{aligned}$$

- We can also calculate its *odd periodic extension* $F(x)$, such that

$$\begin{aligned} F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\ F(x) &= -f(-x) \quad \text{for} \quad x \in (-L, 0) \\ F(x + 2L) &= F(x) \end{aligned}$$

Exercise 4.7: Find the Fourier series of the even periodic extension of

$$f(t) = \begin{cases} t, & 0 \leqslant t \leqslant 1/2 \\ 0, & 1/2 < t \leqslant 1 \end{cases}$$

The even periodic extension of $f(t)$ is $F(t)$, such that

$$\begin{aligned} F(t) &= 0, \quad -1 < t < -1/2 \\ F(t) &= -t, \quad -1/2 \leqslant t < 0 \\ F(t) &= t, \quad 0 \leqslant t \leqslant 1/2 \\ F(t) &= 0, \quad 1/2 < t \leqslant 1 \end{aligned}$$

and $F(t+2) = F(t)$

To calculate its Fourier series

$$a_0 = 2\int_0^{1/2} t\ \mathrm{d}t = \frac{1}{4}$$

$$a_n = 2\int_0^{1/2} t\cos n\pi t\,\mathrm{d}t = 2\left[\frac{t}{n\pi}\sin n\pi t\right]_0^{1/2} - 2\int_0^{1/2}\frac{1}{n\pi}\sin n\pi t\,\mathrm{d}t$$

$$= \frac{1}{n\pi}\sin\frac{n\pi}{2} + 2\left[\frac{1}{n^2\pi^2}\cos n\pi t\right]_0^{1/2} = \frac{1}{n\pi}\sin\frac{n\pi}{2} + \frac{2}{n^2\pi^2}\cos\frac{n\pi}{2} - \frac{2}{n^2\pi^2}$$

The sine and cosine terms will all oscillate between 0, 1 and -1 so lets substitute even and odd values. Let $n = 2m$ then the sine terms will be zero and we have

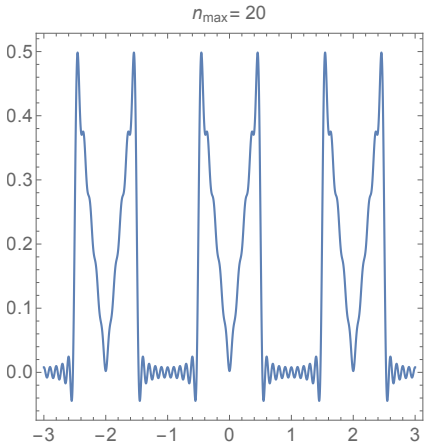
$a_{2m} = \dfrac{2(-1)^m}{(2m)^2\pi^2} - \dfrac{2}{(2m)^2\pi^2}$. For $m = 2l$, $a_{4l} = 0$ and for $m = 2l-1$, $a_{4l-2} = -\dfrac{1}{(2l-1)^2\pi^2}$.

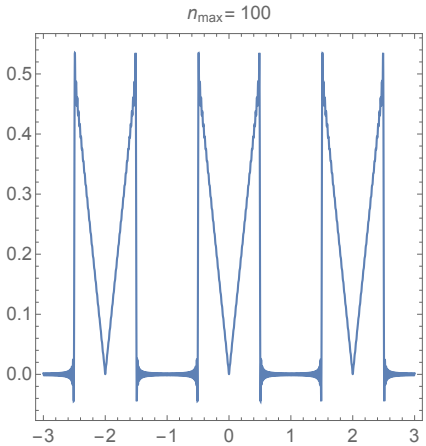
For $n = 2m-1$ the cosine terms are zero and $a_{2m-1} = -\dfrac{(-1)^m}{(2m-1)\pi} - \dfrac{2}{(2m-1)^2\pi^2}$

Therefore,

$$F(t) = \frac{1}{8} - \sum_{m=1}^{\infty}\left(\frac{(-1)^m}{(2m-1)\pi} + \frac{2}{(2m-1)^2\pi^2}\right)\cos(2m-1)\pi t - \sum_{l=1}^{\infty}\frac{1}{(2l-1)^2\pi^2}\cos(4l-2)\pi t$$

where it is continuous.


n_{max}= 20
0.5
0.4
0.3
0.2
0.1
0.0
−3
−2
−1
0
1
2
3


n_{max}= 100
0.5
0.4
0.3
0.2
0.1
0.0
−3
−2
−1
0
1
2
3

Exercise 4.8 (not covered in lecture): Evaluate the series $\displaystyle\sum_{m=1}^{\infty} \frac{1}{(2m-1)^2}$

From Ex 4.7, where F cts.

$$F(t) = \frac{1}{8} - \sum_{m=1}^{\infty} \left(\frac{(-1)^m}{(2m-1)\pi} + \frac{2}{(2m-1)^2\pi^2} \right) \cos(2m-1)\pi t - \sum_{l=1}^{\infty} \frac{1}{(2l-1)^2\pi^2} \cos(4l-2)\pi t$$

Let $t = 0.5$, then

$$\cos(2m-1)\pi t = 0 \quad \text{and} \quad \cos(4l-2)\pi t = -1$$

The mid-point of f (or F) at $t = 0.5$ is 0.25, and thus

$$\frac{1}{4} = \frac{1}{8} + \sum_{l=1}^{\infty} \frac{1}{(2l-1)^2\pi^2} \qquad \Rightarrow \qquad \sum_{l=1}^{\infty} \frac{1}{(2l-1)^2} = \frac{\pi^2}{8}$$